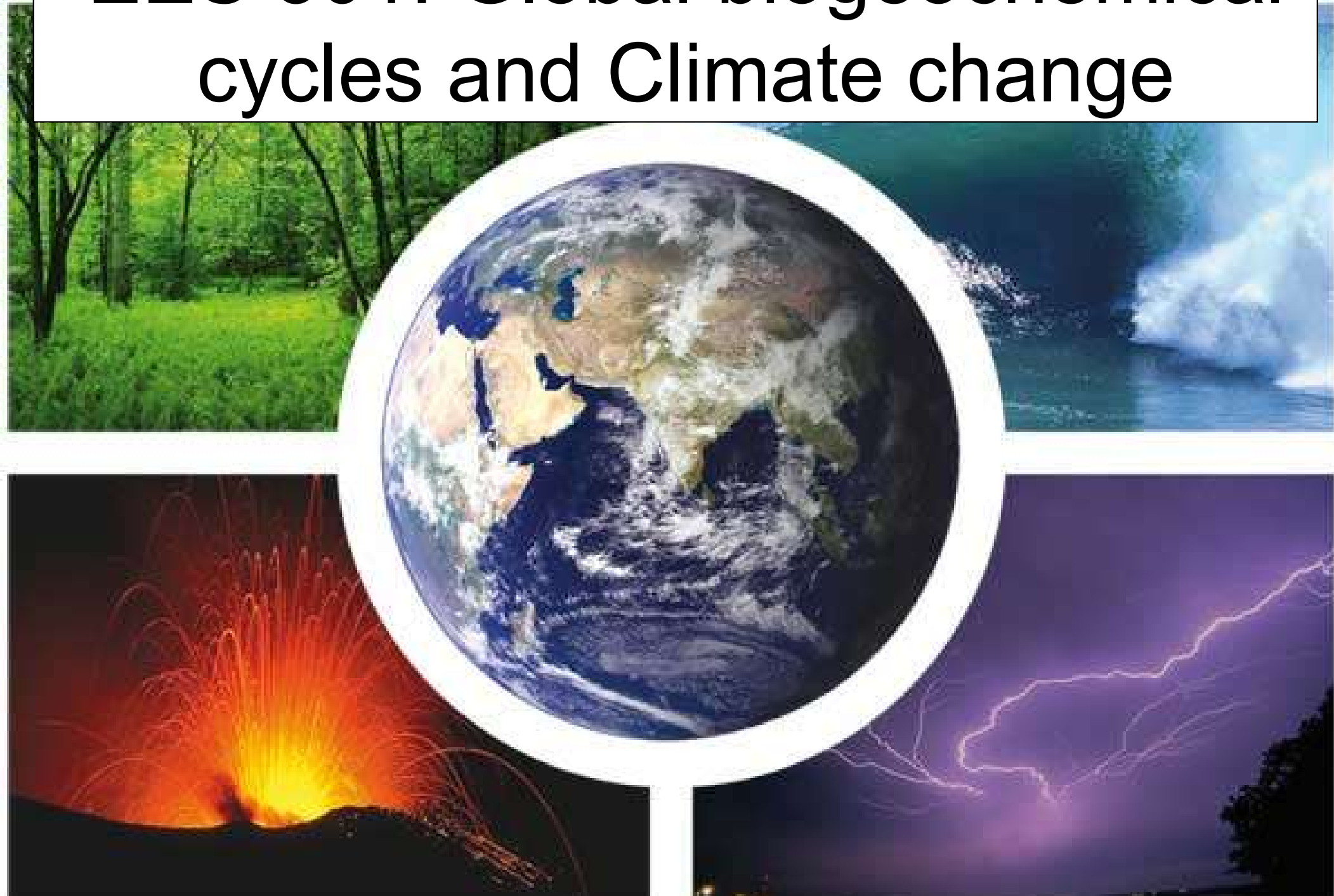


EES 301: Global biogeochemical cycles and Climate change



Atmospheric Transport and Dispersion

Horizontal motion of the atmosphere (wind) is characterized by four spatial scales

Micrometeorology: describes the turbulent dispersion of material on the 1 to 10 km scale. These processes take place on time scales of less than an hour.

Meso-meteorological scale: describes turbulent dispersion and advection on scales of 10 km to a few 100 km. These processes take place within a single day.

Synoptic scale meteorology: Describes the movement of weather systems across 100s to 1000s of km. Advection is the dominant transport process at this scale. Synoptic scale transport takes several days.

Global scale: The large atmospheric circulation cells operate on global scales. These systems undergo long-term changes as a result of climate change.

Micro scale turbulent diffusion and plume transport

A plume spreading from the source (e.g. a stack) in the downwind direction results from advection by the wind. Spreading at a right angle from the wind from turbulence. Turbulent diffusion does not move things by more than a few 10s of km in 2 weeks as ship plume tracks across the Pacific demonstrate.

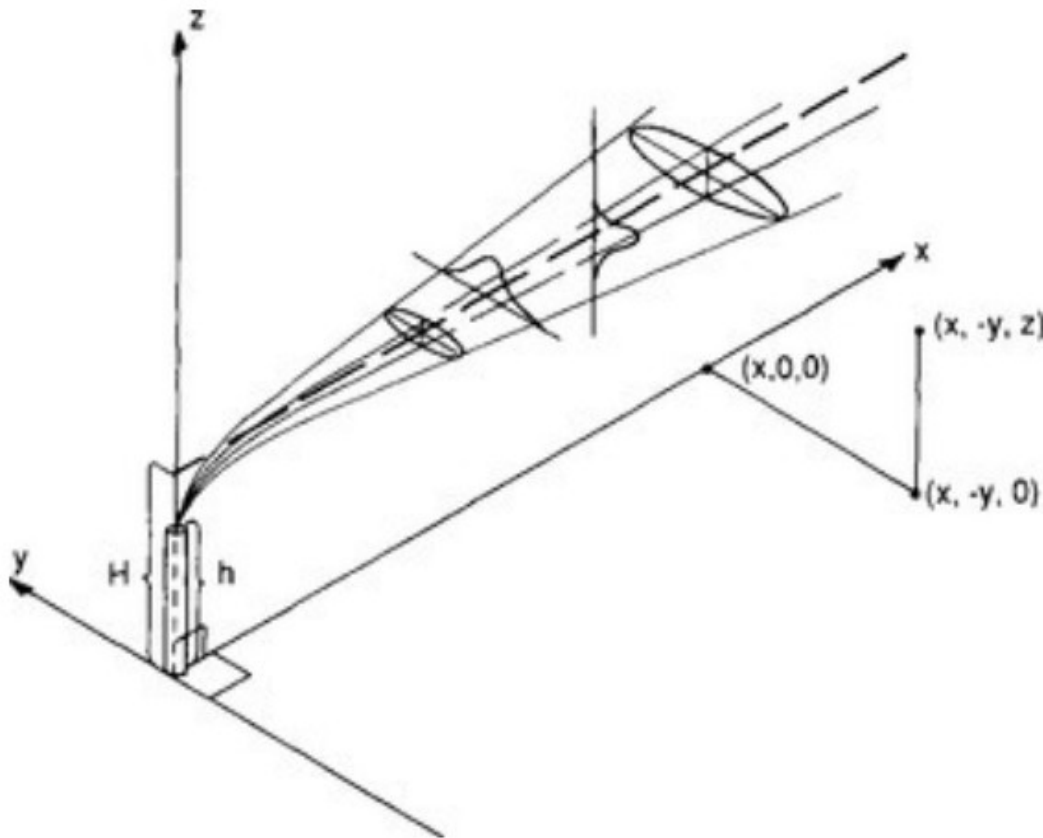
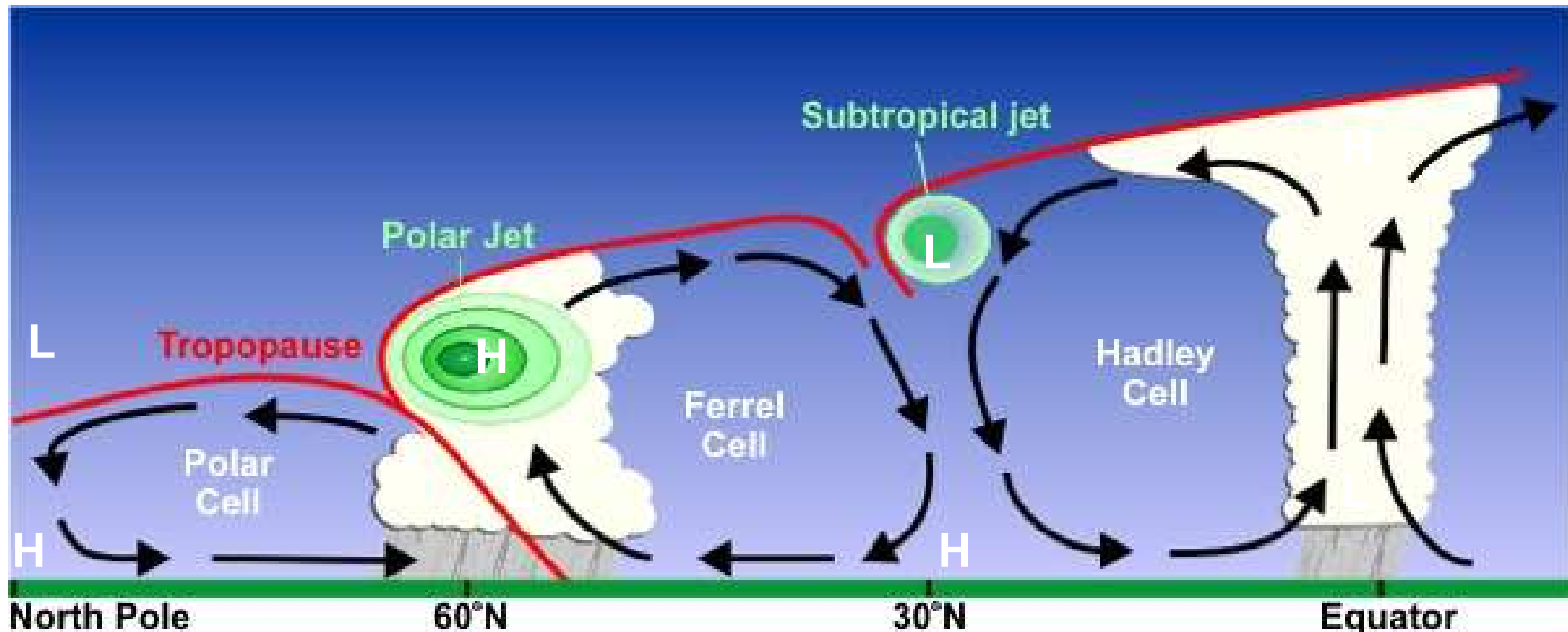


FIG. 7-3 Coordinate system showing the formation downwind from a source of Gaussian distributions of chemical concentrations in the horizontal and vertical. Ellipses denote the loci of two standard deviations.

The General circulation



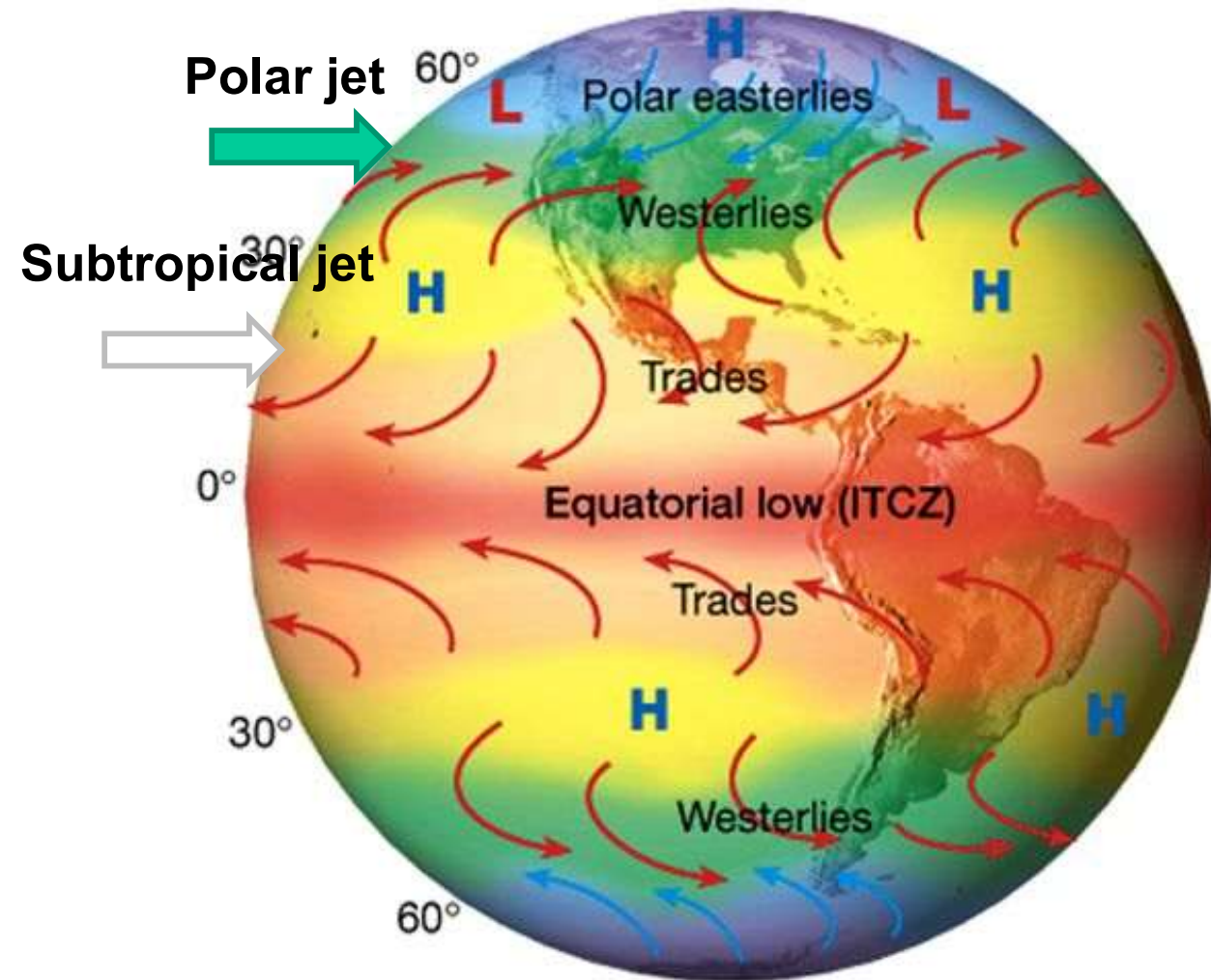
Polar Cell
(thermal):
Driven by heating
at 50 degree
latitude and
cooling at the
poles

Ferrel Cell
(dynamical):
Dynamical response
to Hadley and polar
cells

Hadley Cell (thermal):
*Heating in tropics →
forms surface low and
upper level high → air
converges equatorward
at surface, rises, and
diverges poleward aloft
→ descends in the
subtropics*

The General circulation

Surface winds are easterly in the Hadley cell (Trade winds), and in the Polar cell (Polar easterlies). The surface winds in the Ferrel cell are westerlies.



The real Earth with disruptions of the zonal pattern caused by large landmasses. These disruptions break up pressure zones into semi-permanent high and low pressure cells.

The green area is the zone ^(b) where under normal conditions high pressure and low pressure systems alternate

The semi-permanent cells move creating synoptic scale transport

Synoptic scale transport

The motion of substances on the synoptic scale is often assumed to be pure advection. The flux through a unit area perpendicular to the wind is the product of wind velocity and concentration. $F = c V$

Horizontal motion of the atmosphere or wind is a response of the air to forces including the pressure gradient, the Coriolis force associated with the rotation of the earth and frictional forces acting to retard any motion.

The Geostrophic wind with small frictional effects and acceleration is described as:

$$V_g = \frac{1}{\rho f} \cdot \frac{dP}{dx}$$

with $f = 2\omega \sin\phi$ where ω is the angular velocity due to the rotation of the earth and ϕ is the latitude. The air moves parallel to the isobars (lines of constant pressure). The geostrophic wind blows counterclockwise around low-pressure systems in the northern hemisphere, clockwise in the southern hemisphere and is a good approximation for horizontal wind flow in the upper atmosphere. Near the ground friction and topography can trigger significant deviations.

Synoptic scale transport

The motion in the mid latitudes where high and low pressure areas meet is characterized by “large-scale eddy transport”. Here the eddies are high and low pressure systems which mix air from the polar regions with air from nearer the equator. When cold and dry air is brought into contact with hot and moist air clouds and precipitation occur and a frontal system is

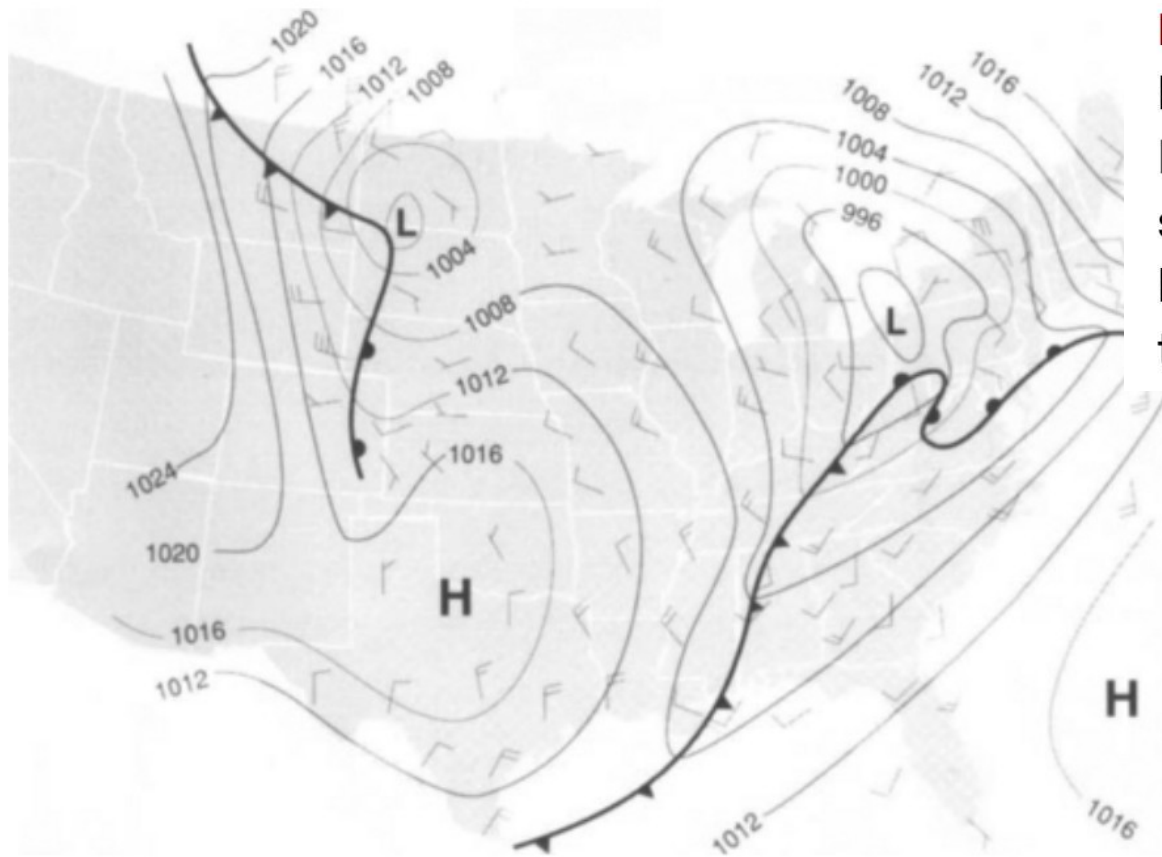
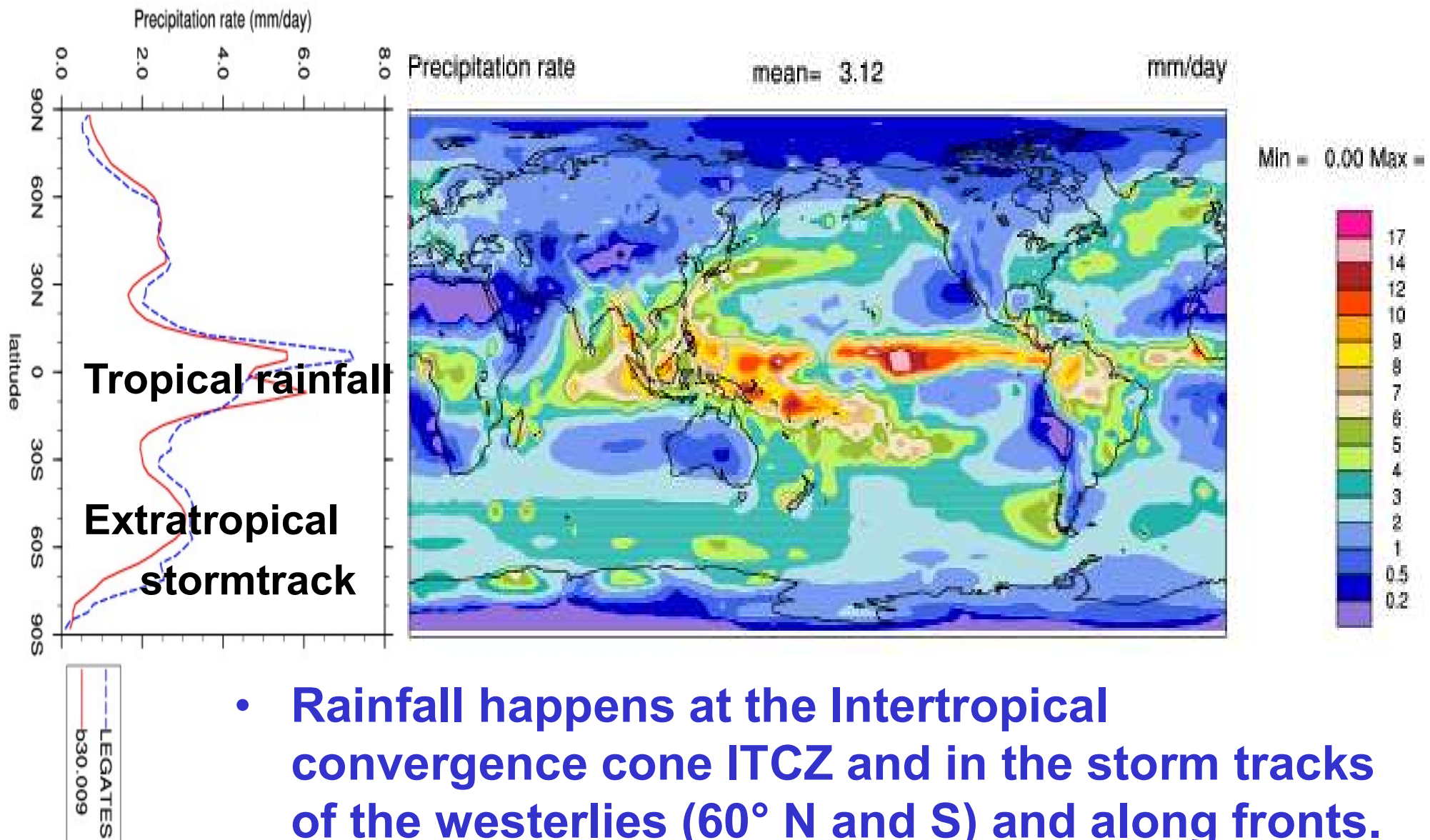


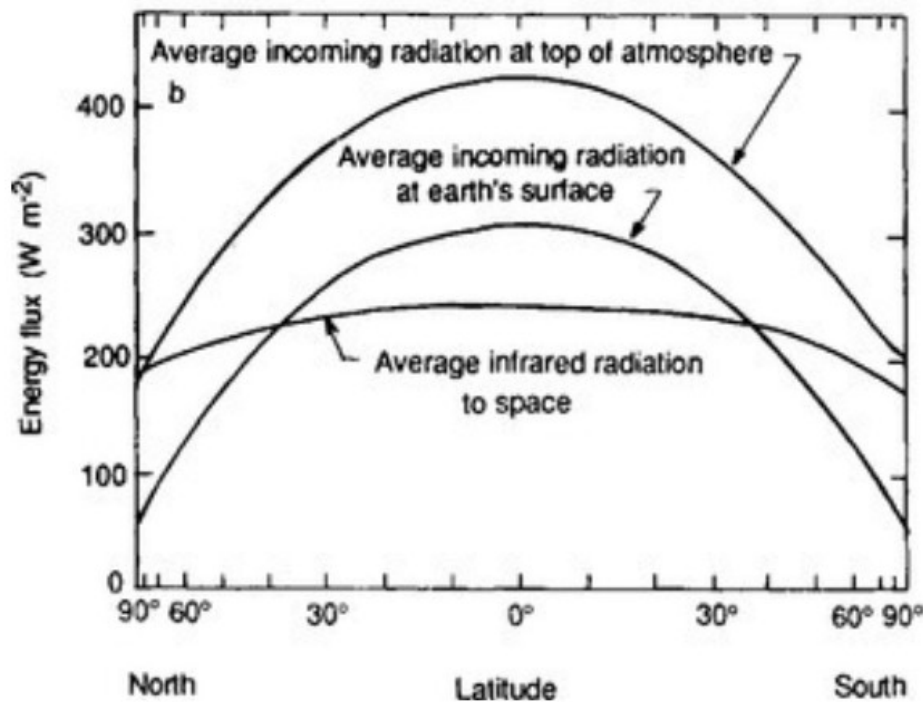
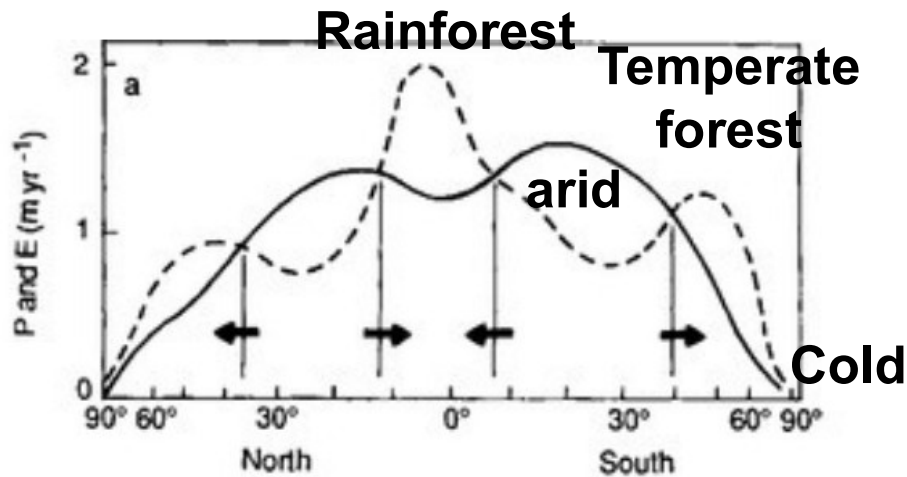
FIG. 7-5 Surface pressure map (milli-bars). Fronts are shown by heavy lines. H = high-pressure system; L = low-pressure system. Wind directions are shown by arrows; wind speeds correspond to the number of bars on the arrow tails.

Barb	
	calm
	5 knots
	10 knots
	50 knots

Most precipitation occurs where atmospheric cells move the air upwards



Climate zones



Climate zones define what can grow where depending on temperature, and precipitation in relation to potential evapotranspiration. They depend on the Earth's average temperature and on the location of land masses relative to atmospheric circulation cells.

FIG. 7-7 (a) Average annual precipitation (P) and evaporation (E) per unit area versus latitude. Arrows represent the sense of the required water vapor flux in the atmosphere. (b) Incoming solar energy (top of atmosphere and surface) and outgoing terrestrial energy versus latitude.

Climate zones

Table 7-1

Water residence time over semi arid and arid zones is higher.

Average residence time of water vapor in the atmosphere as function of latitude

	Latitude range (degrees)								
	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
Average precipitable water (g/cm ²)	4.1	3.5	2.7	2.1	1.6	1.3	1.0	0.7 ^a	0.45 ^a
Average precipitation (g/(cm ² yr))	186	114	82	89	91	77	42	19	11
Residence time (days)	8.1	11.2	12.0	8.7	6.4	6.2	8.7	(13.4)	(15.0)

^aValues extrapolated.

Trace gases in the atmosphere

Table 7-3

Gaseous atmospheric sulfur compounds

Species	Concentration	Sources	Sinks
SO ₂	0–0.5 ppmv (urban) 20–200 pptv (remote)	Oxidation of fossil fuel S Oxidation of S gases, volcanoes	Direct reaction with earth surface, oxidation to sulfate
H ₂ S	0–40 pptv	Biological decay of protein in anaerobic water	Oxidation to SO ₂
CH ₃ SH	Sub-ppbv	Plankton	Oxidation to SO ₂
CH ₃ CH ₂ SH	Sub-ppbv		Oxidation to SO ₂
OCS	500 pptv		Destruction in the stratosphere
CH ₃ SCH ₃	20–200 pptv	Oceanic phytoplankton and algae	Oxidation to SO ₂
CH ₃ SSCH ₃	Small		Oxidation to SO ₂
CS ₂	10–20 pptv		Destruction in the stratosphere and tropospheric OH

Table 7-4

Oxyacids, their salts, and ionized forms that could exist in atmospheric aerosol particles

Oxyacid	Formula	form
Sulfuric	H_2SO_4	HSO_4^- , SO_4^{2-} NH_4HSO_4 $(\text{NH}_4)_3\text{H}(\text{SO}_4)_2$ $(\text{NH}_4)_2\text{SO}_4$ MgSO_4 CaSO_4 Na_2SO_4 $\text{R}-\text{O}-\text{SO}_3^{2-}$
Sulfurous	$\text{SO}_2 \cdot \text{H}_2\text{O}$	HSO_3^- , SO_3^{2-}
Sulfonic	$\text{R}-\text{SO}_3-\text{H}$	$\text{R}-\text{SO}_3^-$
Hydroxymethane sulfonic	$\text{CH}_2(\text{OH})\text{SO}_3\text{H}$	$\text{CH}_2(\text{OH})\text{SO}_3^-$
Dithionic	$\text{H}_2\text{S}_2\text{O}_6$	$\text{S}_2\text{O}_6^{2-}$
Thiosulfuric	$\text{H}_2\text{S}_2\text{O}_3$	$\text{S}_2\text{O}_3^{2-}$

SO_2 is oxidized to sulfuric acid H_2SO_4 ($K_1=1.7 \times 10^{-2}$, $K_2=6.2 \times 10^{-8}$) and sulfate salts.
 SO_2 emissions are the main contributors to acid rain.

The main removal mechanism of sulfate is wet deposition (rainout). The amount of solute removed via CCN is:

$$[X] = \frac{\varepsilon(X)_{\text{air}}}{M_X L}$$

Where $[X]$ is the average molarity of the solute X in cloud water, ε the fraction of the aerosol particles of X that are activated CCN, M_X the molecular weight of X , $(X)_{\text{air}}$ the concentration of X in the air entering the cloud (g/m^3) and L the liquid water content of the cloud (L/m^3)

Trace gases in the atmosphere

Table 7-5

Nitrogen atmospheric compounds

Species	Concentration	Source	Sink
NH ₃	0–20 ppbv	Biological	Precipitation
RNH ₂ ... R ₃ N	— ^a	Biological	Precipitation
N ₂	78.084%	Primitive volatile, denitrification	Biological nitrification
N ₂ O	~ 0.3 ppmv	Biological	Photolysis in the stratosphere
(N ₂ O ₃)	— ^a	Reaction intermediate	— ^a
NO	0–0.1 ppmv	Oxidation of N ₂ in combustion	HNO ₃
NO ₂	0–0.1 ppmv	NO oxidation	HNO ₃
HNO ₂	— ^a	OH· + NO	Precipitation
HNO ₃	— ^a	OH· + NO ₂	Precipitation

^aGlobal values not established.

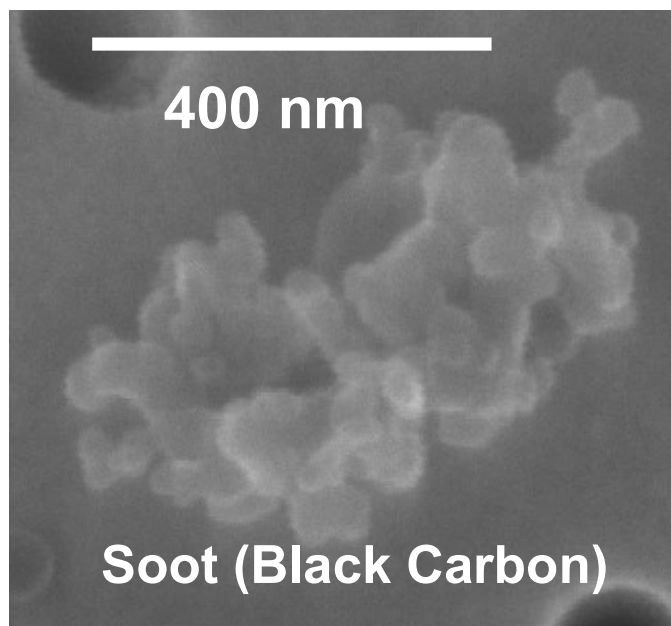
Reactive nitrogen in aerosols

Table 7-6

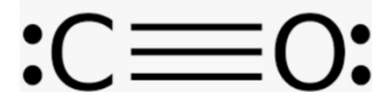
Nitrogenous aerosol constituents

Species	Concentration
$(\text{NH}_4)_2\text{SO}_4$	Up to $30 \mu\text{g}/\text{m}^3$
NH_4HSO_4	
Amine sulfates	—
NH_4NO_3	Up to a few $\mu\text{g}/\text{m}^3$, but only in polluted air
Other nitrates	—
Organic N	—

Carbon compounds in the atmosphere



CO source combustion, sink:
 $\text{CO} + \text{OH}\cdot \rightarrow \text{CO}_2 + \text{H}\cdot$



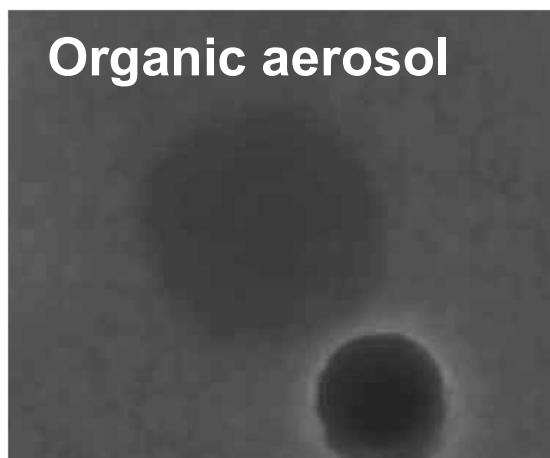
CO₂ source combustion, respiration, oxidation of CO & VOCs. Sink Dissolution in aqueous media to form carbonic acid.



$$P(\text{CO}_2) = K_H[\text{H}_2\text{CO}_3^*]$$

$$K_H = 29 \text{ atm mol}^{-1} \text{ L}$$

$$\text{H}_2\text{CO}_3^* = \text{H}_2\text{CO}_3 + \text{CO}_2(\text{aq})$$



If you have to calculate M of H_2CO_3^* for water in equilibrium with a partial pressure of 300 ppm CO₂ which is the same as 300×10^{-6} atm then:
 $300 \times 10^{-6} \text{ atm} / (29 \text{ atm mol}^{-1} \text{ L})$
 $= 1 \times 10^{-5} \text{ mol/L}$

If you have to calculate pH then you need the $K_1 = 4.5 \times 10^{-7} \text{ mol/L}$
 $\text{H}_2\text{CO}_3 \rightleftharpoons \text{HCO}_3^- + \text{H}^+$
 $K_1 = 4.5 \times 10^{-7} = [\text{HCO}_3^-][\text{H}^+] / [\text{H}_2\text{CO}_3]$
 $= [\text{HCO}_3^-]^2 / 1 \times 10^{-5}$
 $[\text{HCO}_3^-] = [\text{H}^+] = 2.1 \times 10^{-6} \text{ M/L};$
 $\text{pH} = -\log_{10}(a_{\text{H}^+}) \approx -\log_{10}([\text{H}^+]/\text{M})$
 $\text{pH} = 5.6$

Organic carbon

Table 7-7

Atmospheric hydrocarbons

Class	Compound	Typical source	Probable sink	Conc.
Alkanes	Methane	Microbes natural gas	OH	1.8 ppmv
	Ethane	Auto	OH	0-100 ppbv
	Hexane	Auto	OH	0-30 ppbv
	(Up to C ₃₇)	—	—	—
Alkenes	Ethene	Auto, microb., vegetation	OH, O ₃	1-1000 ppbv
	Isoprene	Trees, auto	OH	0.2-30 ppbv
Alkynes	Acetylene	Auto, microbes	OH	0.2-200 ppbv
Terpenes	α -Pinene	Trees	O ₃ , OH	0-1 ppbv
	Limonene	Trees	O ₃ , OH	0-1 ppbv
Cyclic hydrocarbons	Cyclopentane	Auto	OH	0-10 ppbv
	Cyclohexane	Auto	OH	0-10 ppbv
	Cyclopentene	Auto	O ₃ , OH	0-10 ppbv
Aromatic hydrocarbons	Benzene	Auto	OH, O	—
	Toluene	Auto	OH	0-100 ppbv
Polynuclear aromatic hydrocarbons	Phenanthrene	Aluminum manufacture, combustion		0-300 ng/m ³
	Benzo(a)pyrene	Auto, wood combustion		0-100 ng/m ³

Organic carbon

Table 7-8

Oxygenated organic compounds

Class	Example	Typical source	Probable sink	Conc.
Aliphatic aldehydes	Formaldehyde		<i>hν</i> , OH, rain	1–100 ppbv
	Acetaldehyde	Animal waste	<i>hν</i> , OH	1–10 ppbv
Olefinic aldehydes	Acrolein	Auto	O ₃	0–1 ppbv
Aromatic aldehydes	4-methyl-benzaldehyde	Auto	—	0–300 pptv
Aliphatic ketones	Acetone	Animal waste	<i>hν</i>	0–10 ppbv
Cyclic ketones	1,2-cyclopentane	Known plant	—	No data
Aromatic ketones	Acetophone	Auto plant emissions	—	No data
Aliphatic acids	Formic acid	—	Rain	0–100 ppbv
	Pentanedioic acid	Cyclopentene	Rain	0–1 mg/m ³
Cyclic acids	Pinonic acid	α-Pinene + O ₃	Rain	
Aromatic acids	Benzoic acid	Auto	Rain	0–400 ng/m ³
Aliphatic alcohols	Methanol CH ₃ OH	Animal waste	—	0–100 ppbv
Phenols	Phenol	Auto	—	0–3 ppbv

Organic carbon

Table 7-9

Atmospheric nitrogen-containing organic compounds

Class	Example	Typical source	Probable sink	Conc.
Amines	Methylamine	Protein decay	H ₂ SO ₄ aerosol	—
Nitrates	Peroxy acetyl nitrate	Photochemical	—	0–100 ppbv
Heterocyclic N compounds	Benzo[f]quinoline	Coal combustion	—	0–200 pg/m ³

Table 7-10

Atmospheric sulfur-containing organic compounds

Class	Example	Typical source	Probable sink	Conc.
Mercaptans	Methyl mercaptan	Microbiota	Oxidation to SO ₂ by OH	0–4 ppbv
Sulfides	Dimethyl sulfide	Algae	Oxidation to SO ₂ by OH	20–300 pptv

Organic carbon

Table 7-11

Atmospheric halogen-containing organic molecules

Class	Example	Typical source	Probable sink	Conc.
Halogenated aliphatic compounds	CH ₃ Cl	Biological	OH	620 pptv
	CH ₂ Cl ₂	Industrial	OH	30 pptv
	CH ₃ CCl ₃	Solvent	—	140 pptv
	CCl ₄	Solvent	—	130 pptv
	CFCl ₃ (CFC-11)	Propellant, refrigerant	Stratosphere	220 pptv
	CF ₂ Cl ₂ (CFC-12)	Refrigerant	Stratosphere	375 pptv
Halogenated aromatic compounds	DDT	Pesticide	—	0.009–500 ng/m ³

Aerosol

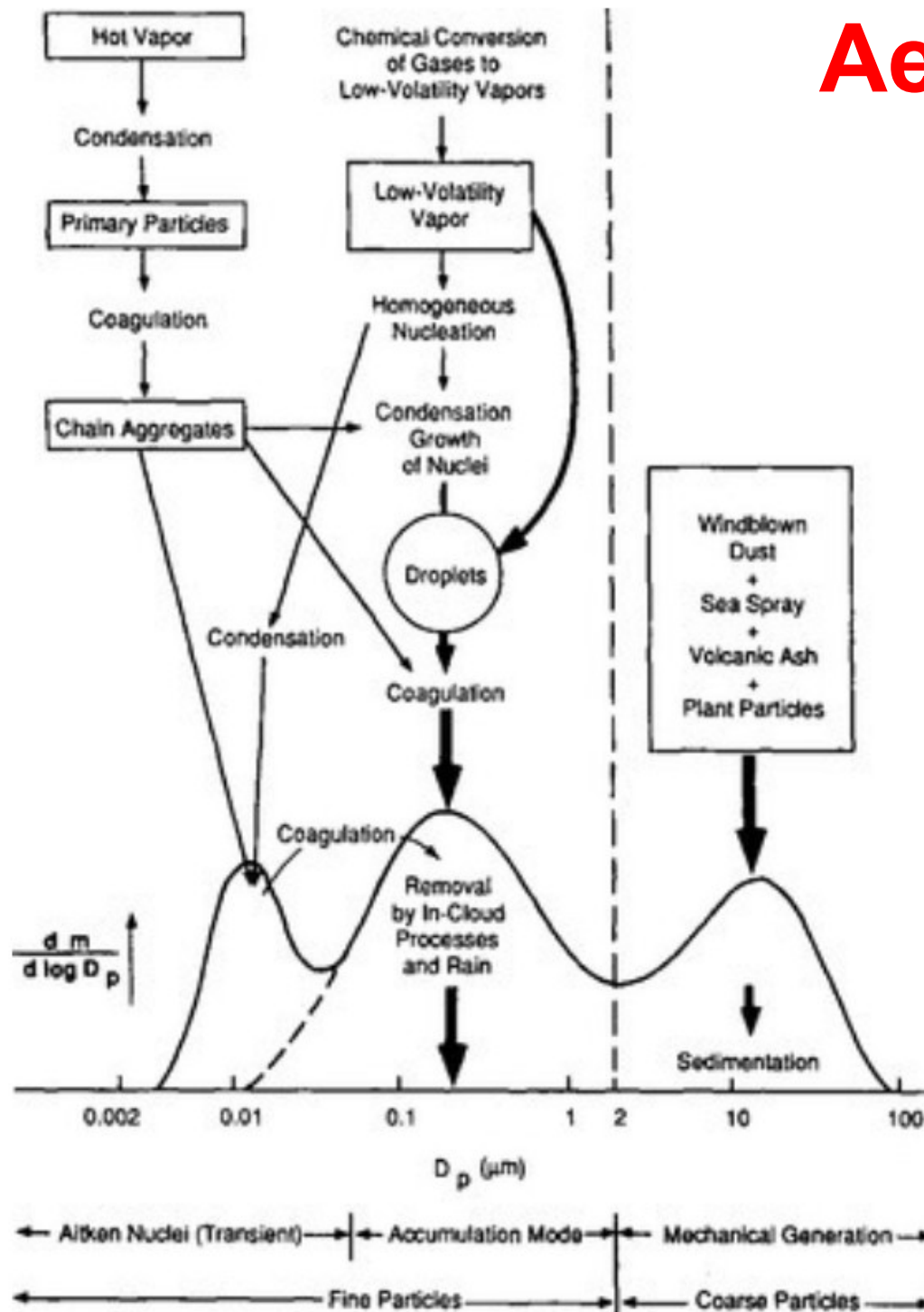


FIG. 7-12 Schematic of an atmospheric aerosol size distribution. This shows the three mass modes, the main sources of mass for each mode, and the principal processes involved in inserting mass into and removing mass from each mode (m = mass concentration, D_p = particle diameter). (Reproduced with permission from K. T. Whitby and G. M. Sverdrup (1983). *California aerosols: their physical and chemical characteristics*. In "The Character and Origin of Smog Aerosols" (G. M. Hidy, P. K. Mueller, D. Grosjean, B. R. Appel, and J. J. Wesolowski, eds), p. 483, John Wiley, New York.)

Aerosol

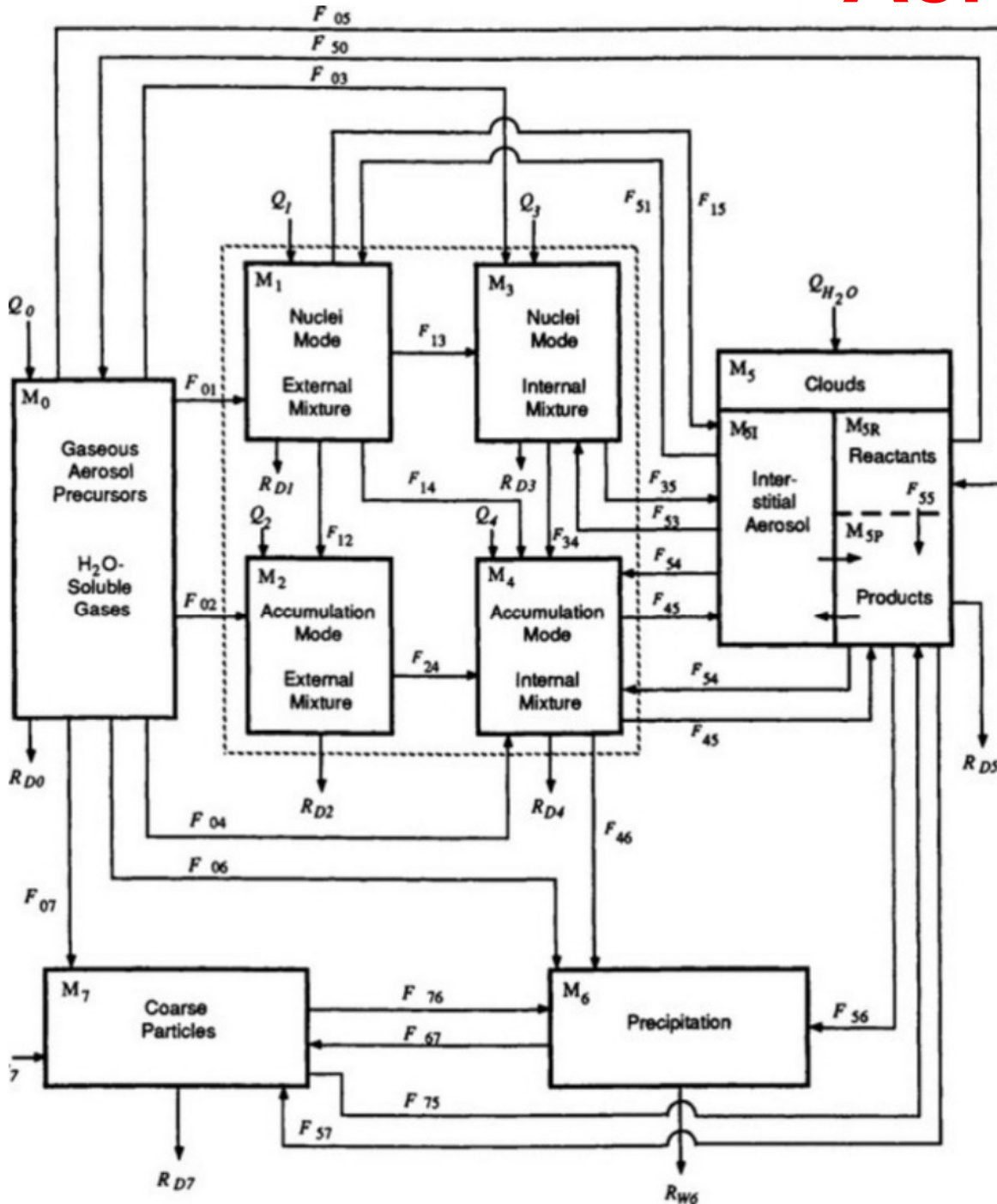


FIG. 7-13 Physical transformations of trace substances in the atmosphere. Each box represents a physically and chemically definable entity. The transformations are given in F_{ij} . The physical transformations are as follows: F_{01} : Production of new nuclei-mode particles; F_{02} : Growth of existing accumulation-mode particles by the deposition of products of chemical reactions; F_{03} : Growth of pre-existing nuclei-mode

PRACTICE QUESTIONS

You can submit the answer to these questions by Monday 15th September if you want to get them corrected before the Midsem exam

MCQ questions

1. What direction does air move according to the pressure gradient force?
 - a) High pressure to low pressure
 - b) Low pressure to high pressure
 - c) Left in the southern hemisphere and right in the northern hemisphere
 - d) Right in the southern hemisphere and left in the northern hemisphere
2. What is the driving force behind the Coriolis effect?
 - a) Rotation of the earth
 - b) Pressure gradients
 - c) Temperature differences
 - d) Atmospheric circulations
3. In what direction do geostrophic winds blow?
 - a) Perpendicular to isobars
 - b) Parallel to isobars
 - c) High pressure to low pressure
 - d) None of these are correct.
4. The geostrophic wind is a good approximation for
 - a) horizontal wind flow in upper atmosphere.
 - b) vertical wind flow in upper atmosphere.
 - c) horizontal wind flow at the earth surface.
 - d) vertical wind flow at the earth surface.

MCQ questions

1. You are standing in the Northern Hemisphere in a field, facing north, with the wind at your back. In which direction is the lowest pressure?
 - a) North (straight ahead)
 - b) South (behind you)
 - c) East (to your right)
 - d) West (to your left)
2. The Coriolis effect is an "apparent" force caused by
 - a) Friction
 - b) rotation of the earth.
 - c) pressure gradient force.
 - d) gravity.
3. The jet stream is located near the tropopause and _____.
 - a) flows west to east in both hemispheres
 - b) flows west to east only in the Northern Hemisphere
 - c) flows west to east only in the Southern Hemisphere
 - d) flows east to west in both hemispheres.
4. The pressure gradient force is
 - a) parallel to isobars.
 - b) directed from high to low pressure.
 - c) directed from low to high pressure.
 - d) always greater than friction.

MCQ questions

1. Which is the major source for sulphur dioxide?
 - a) Volcanic eruptions
 - b) Coal and crude oil combustion
 - c) Burning of petrol
 - d) Sewage treatment process
2. Which is the major source for reduced sulfur compounds (DMS, H_2S) in the atmosphere?
 - a) Oceanic phytoplankton
 - b) Trees
 - c) Anoxic waters
 - d) wetlands
2. What are the largest sources of nitrogen oxide emissions
 - a) Lightening
 - b) Soil microbes
 - c) Fossil fuel combustion
 - d) Forest fires.

Numerical

1. If there is $5 \mu\text{g}/\text{m}^3$ of ammonium sulfate $(\text{NH}_4)_2\text{SO}_4$ aerosol in the air entering a cloud with $1 \text{ mL}/\text{m}^3$ liquid water content and $\epsilon = 1$ then what will the molarity of the cloud water?
2. How many M of H_2CO_3^* will be there in rainwater in equilibrium with 420 ppm CO_2 (present day levels) and 1000 ppm of CO_2 (projected 2100 levels for business as usual). Also calculate the pH of rainwater containing only carbonic acid for both cases. $K_{\text{H}} = 29 \text{ atm mol}^{-1} \text{ L}$, $K_1 = 4.5 \times 10^{-7}$
3. Calculate the pH of rainwater that is in equilibrium with an air that has an SO_2 concentration of 50 ppbv. $K_{\text{H}} = 1.0 \text{ mol L}^{-1} \text{ atm}^{-1}$, $K_1 = 1.7 \times 10^{-2}$.
4. You have measured a pH of 4.2 in rainwater on your campus. Assuming the acidity is from SO_2 then what is the concentration of SO_2 that must be in the air in ppbv
5. Oxygen in air is 21%. Calculate the molar concentration of dissolved oxygen in water that is saturated. $K_{\text{H}} = 1.28 \times 10^{-3} \text{ mol L}^{-1} \text{ atm}^{-1}$

Long answer style

1. Explain how the distribution of major biomes (=climate zones) on earth e.g. tropical rainforests, temperate forest and the arid desert belt best seen in the Sahara and Arabian desert are related to the three large atmospheric circulation cells.
2. Which are the main classes of hydrocarbons that exist in polluted atmospheres
3. Which are aerosols number distributions critical for the hydrological cycle?